Document details

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Issued to	
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Document control

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1	28/08/2024	First version of the document	Lee Sanderson	
2	16/09/2024	Added section for remote access using Rust Desk.	Lee Sanderson	

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Quality Management System Standards

The standard is applicable to the management of activities associated with the design, development, deployment, projects and operational support of mobile wireless data solutions for the transportation sector. These solutions include on-vehicle systems, fixed network systems, backhaul systems, network operating centre services, hardware and software.

Preface

Document Guidelines

The following notices and statements are used in this document:

Draws your attention to particularly important information on handling the product, the product itself or to a particular part of the documentation.
Indicates that if the proper precautions are not taken, this can result in product damage.

Emphasizes or supplements important points of the main text. Helps users apply the techniques and procedures described in the text suggesting alternative methods that may not be obvious.

Documentation Repository

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Reporting issues with this document

While reasonable precautions have been taken, technical inaccuracies or typographical errors may appear in the document.

External document users can report issues through the Nomad Service Desk or through a Nomad Project Engineer.

Contents

1. Introduction

The purpose of this document is to provide instructions as to how to connect physically to the Nomad 5001c CCU using a KVM and additional multi adapters.

This document provides an overview of what hardware and software is required to physically connect to the Nomad 5001c CCU, as well as a clear diagram and step by step instructions to make this connection. Finally the document overviews what software will be required to view the output of the CCU via the KVM.

In addition to this this document shows how to share an internet connection via the ethernet adapter with a Linux Machine.

2. Required Hardware

This section will provide an overview of what hardware is required in order to connect to the Nomad 5001c CCU.

Hardware	Image
USB C to VGA Adapter	
USB Multi-adapter (Expander)	
Startech KVM	

USB to Ethernet Adapter	
Nomad Laptop	

3. Instructions

3.1 CONNECT SERIAL TO 5001C

Step 1 Connect a USB C to VGA adapter to the USB C port.

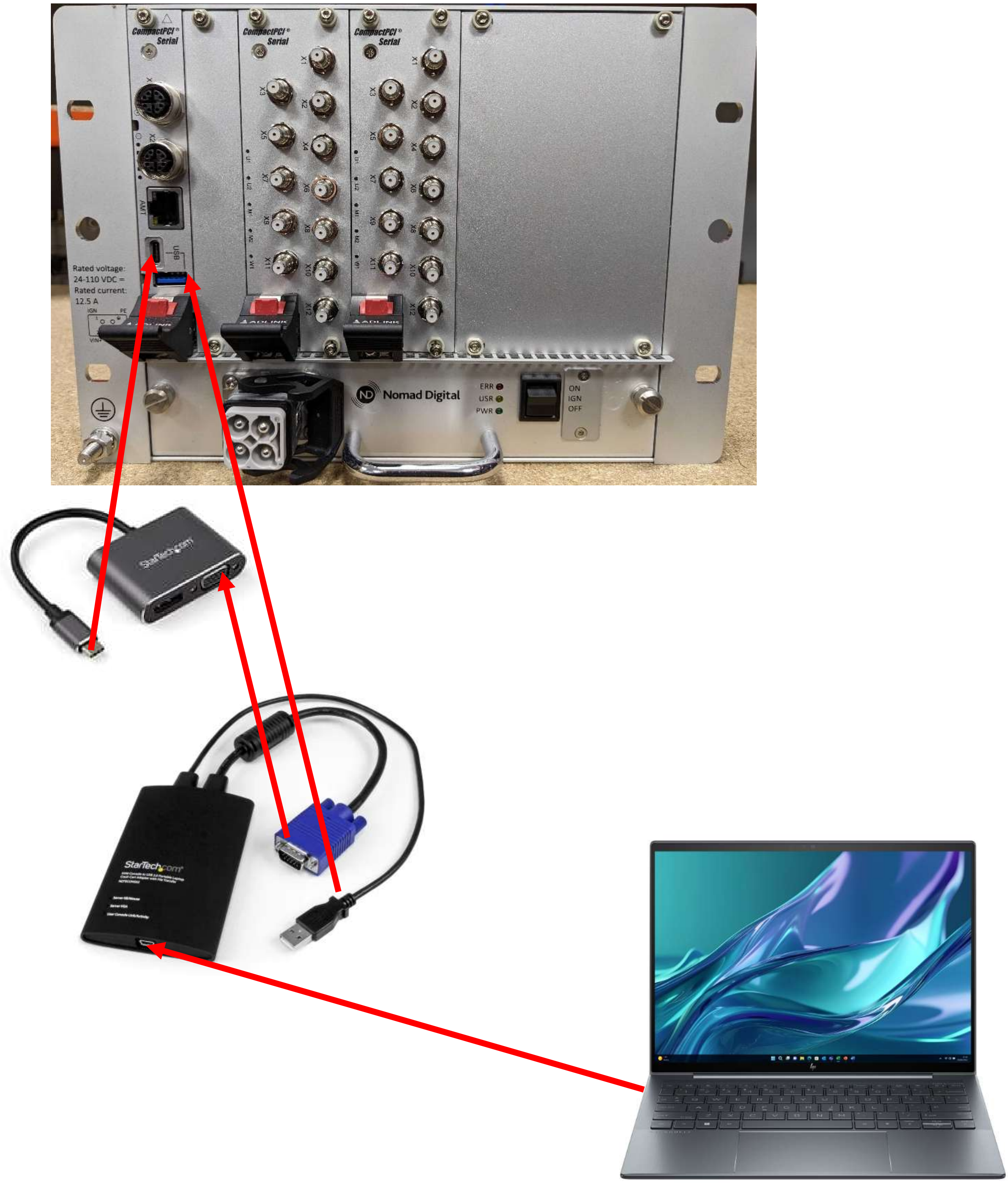
Step 2 Connect a USB Multi Hub to the USB Port.

Step 3 Connect the KVM's VGA cable to USB C to VGA adapter and plug the USB cable into the Multi Hub.

Step 4 Connect Cable from KVM to laptop.

A restart of the CCU may be required for console access.

4. Diagram


CompactPCI® Serial
CompactPCI® Serial
CompactPCI® Serial
Rated voltage:
24-110 VDC =
Rated current:
12.5 A
Nomad Digital
ERR
USR
PWR
ON
IGN
OFF
StarTech.com
StarTech.com

5. Software

To view the output from the KVM once the device is connected you will need an application from the StarTech website called “USB Crash Kart Adapter”

This software can be downloaded from the below link from the StarTech website.

https://www.startech.com/en-gb/support/drivers-and-downloads

Please note Nomad Test Laptops before they are sent already have this software pre-installed.

6. Remote Access (RustDesk)

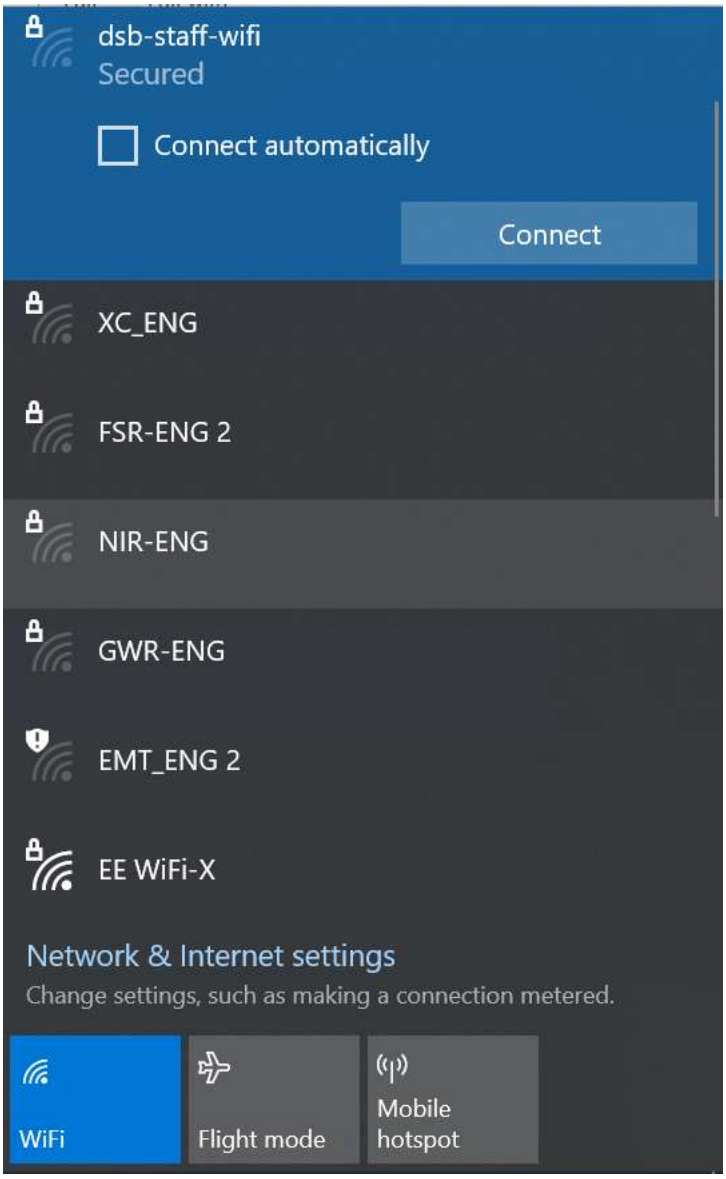
In the case further Nomad support is required Nomad can connect remotely via the Laptop to the CCU once the console cable is connected.

Use the below software for the remote support:

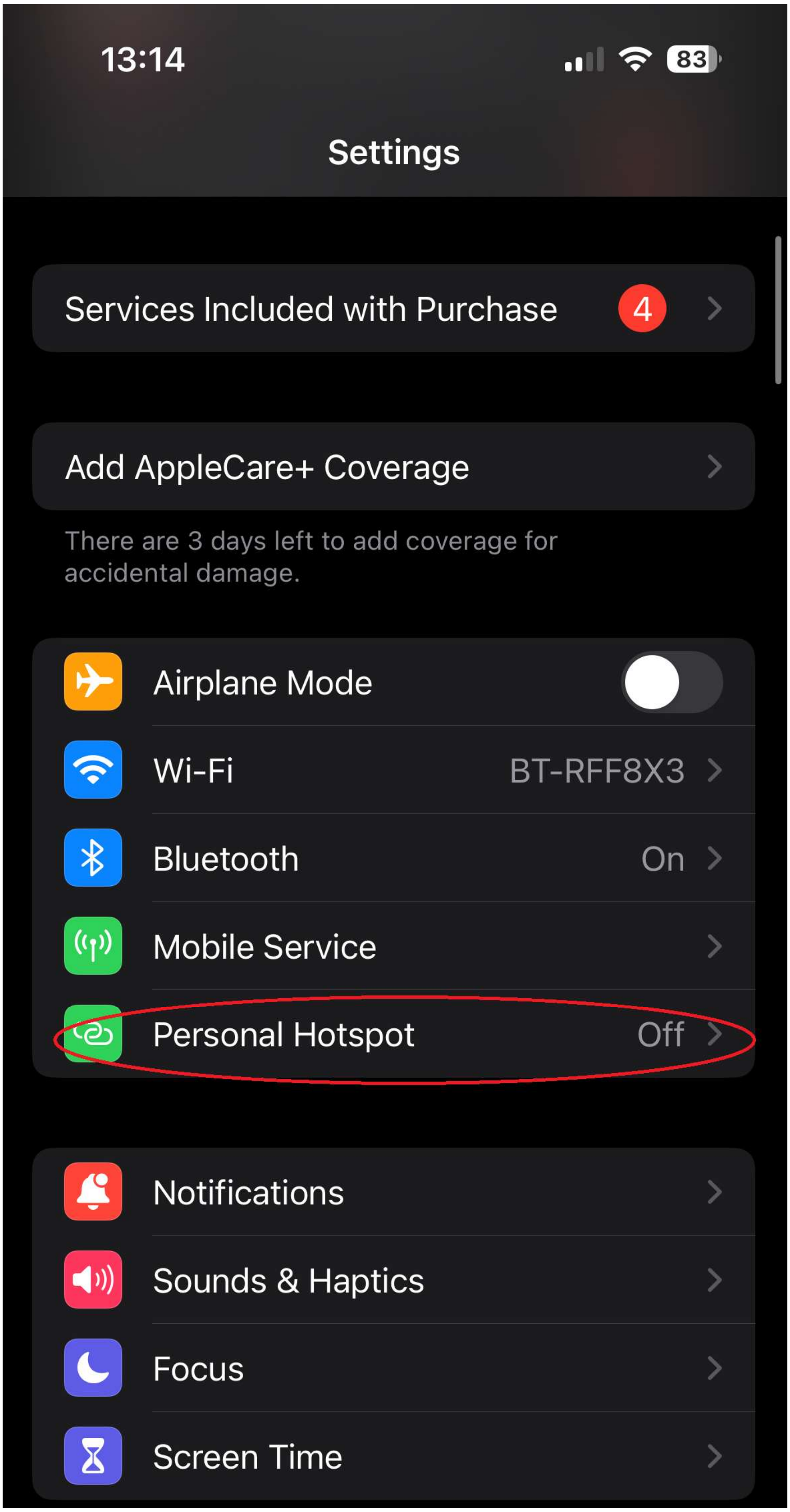
https://rustdesk.com/

Please note Nomad Test Laptops before they are sent already have this software pre-installed.

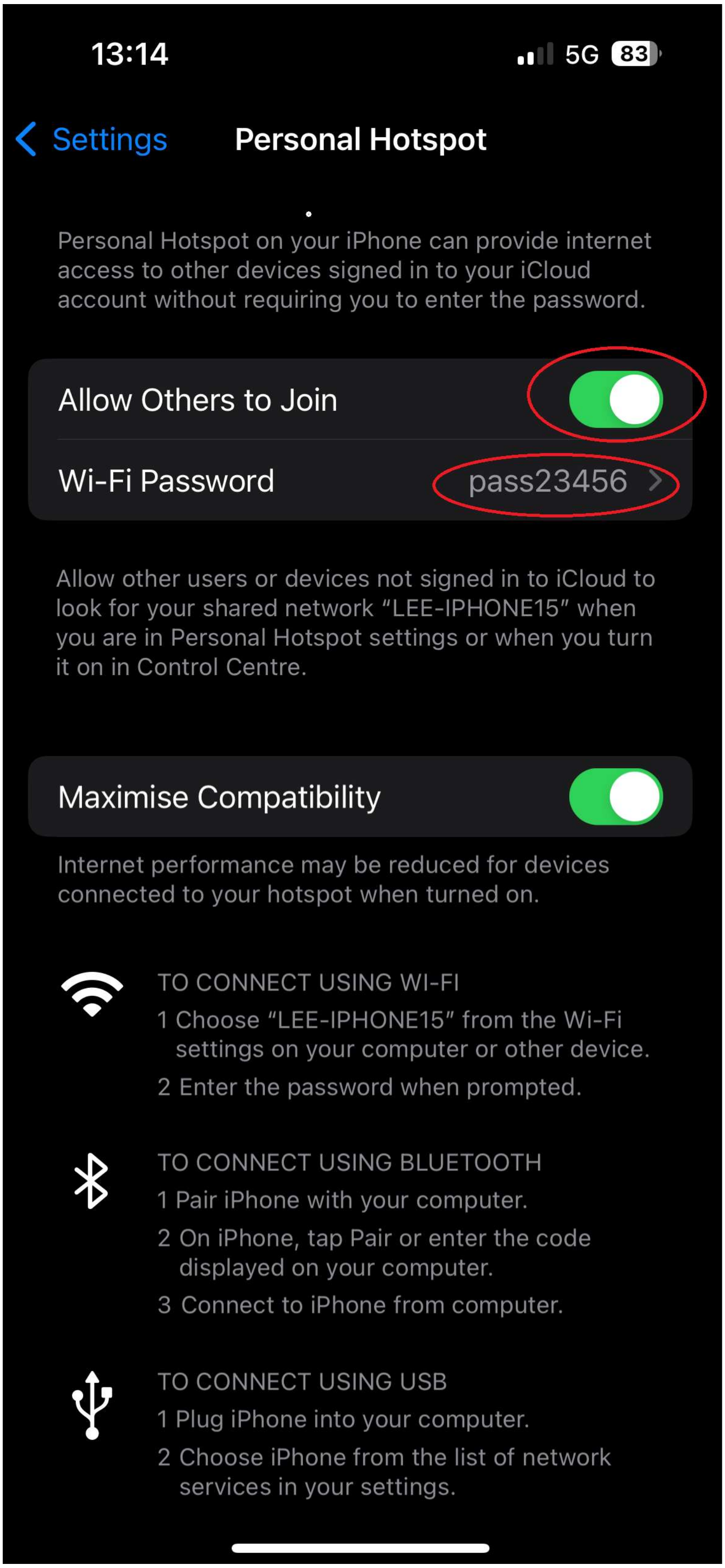
Please ensure the laptop is connected to Wifi to allow remote connection, this can be either the depot/factory Wifi or if not available use hotspot on a mobile device.



Navigate to settings on the mobile phone and select personal hotspot.

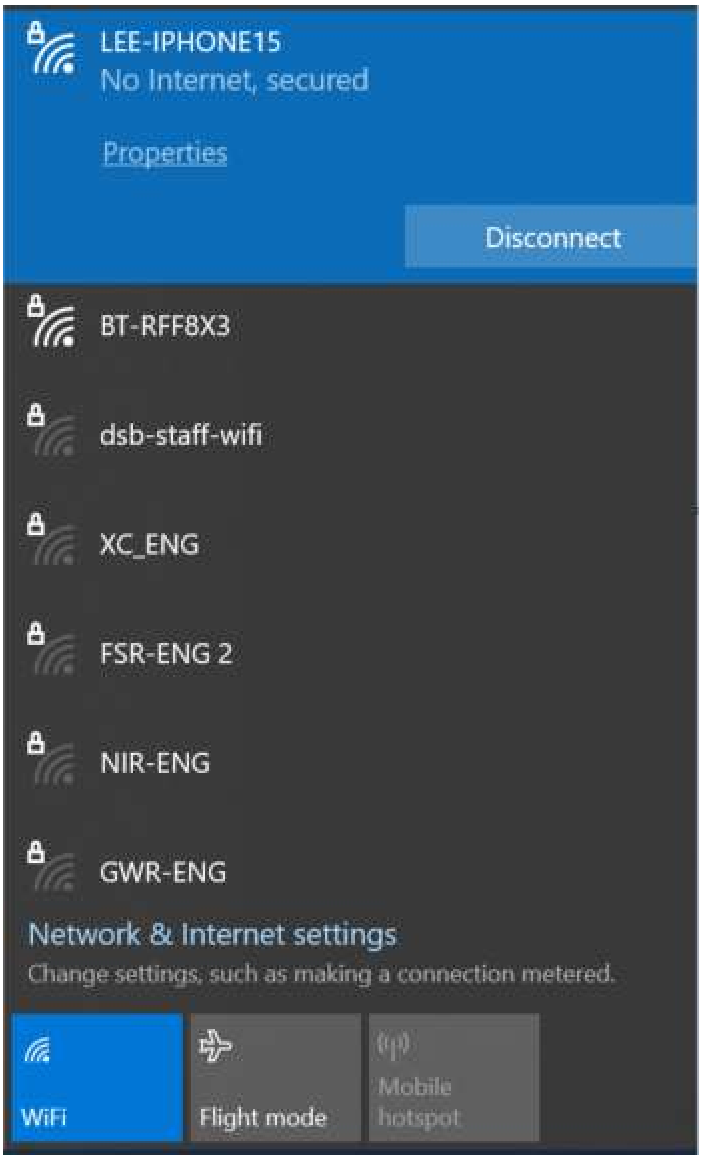

13:14
83
Settings
Services Included with Purchase
4
Add AppleCare+ Coverage
There are 3 days left to add coverage for accidental damage.
Airplane Mode
Wi-Fi
BT-RFF8X3
Bluetooth
On
Mobile Service
Personal Hotspot
Off
Notifications
Sounds & Haptics
Focus
Screen Time

Ensure “allow others to join” option is turned on and a password is set.


13:14
5G 83
Settings
Personal Hotspot
Personal Hotspot on your iPhone can provide internet access to other devices signed in to your iCloud account without requiring you to enter the password.
Allow Others to Join
Wi-Fi Password
pass23456
Allow other users or devices not signed in to iCloud to look for your shared network "LEE-IPHONE15" when you are in Personal Hotspot settings or when you turn it on in Control Centre.
Maximise Compatibility
Internet performance may be reduced for devices connected to your hotspot when turned on.
TO CONNECT USING WI-FI
1 Choose "LEE-IPHONE15" from the Wi-Fi settings on your computer or other device.
2 Enter the password when prompted.
TO CONNECT USING BLUETOOTH
1 Pair iPhone with your computer.
2 On iPhone, tap Pair or enter the code displayed on your computer.
3 Connect to iPhone from computer.
TO CONNECT USING USB
1 Plug iPhone into your computer.
2 Choose iPhone from the list of network services in your settings.

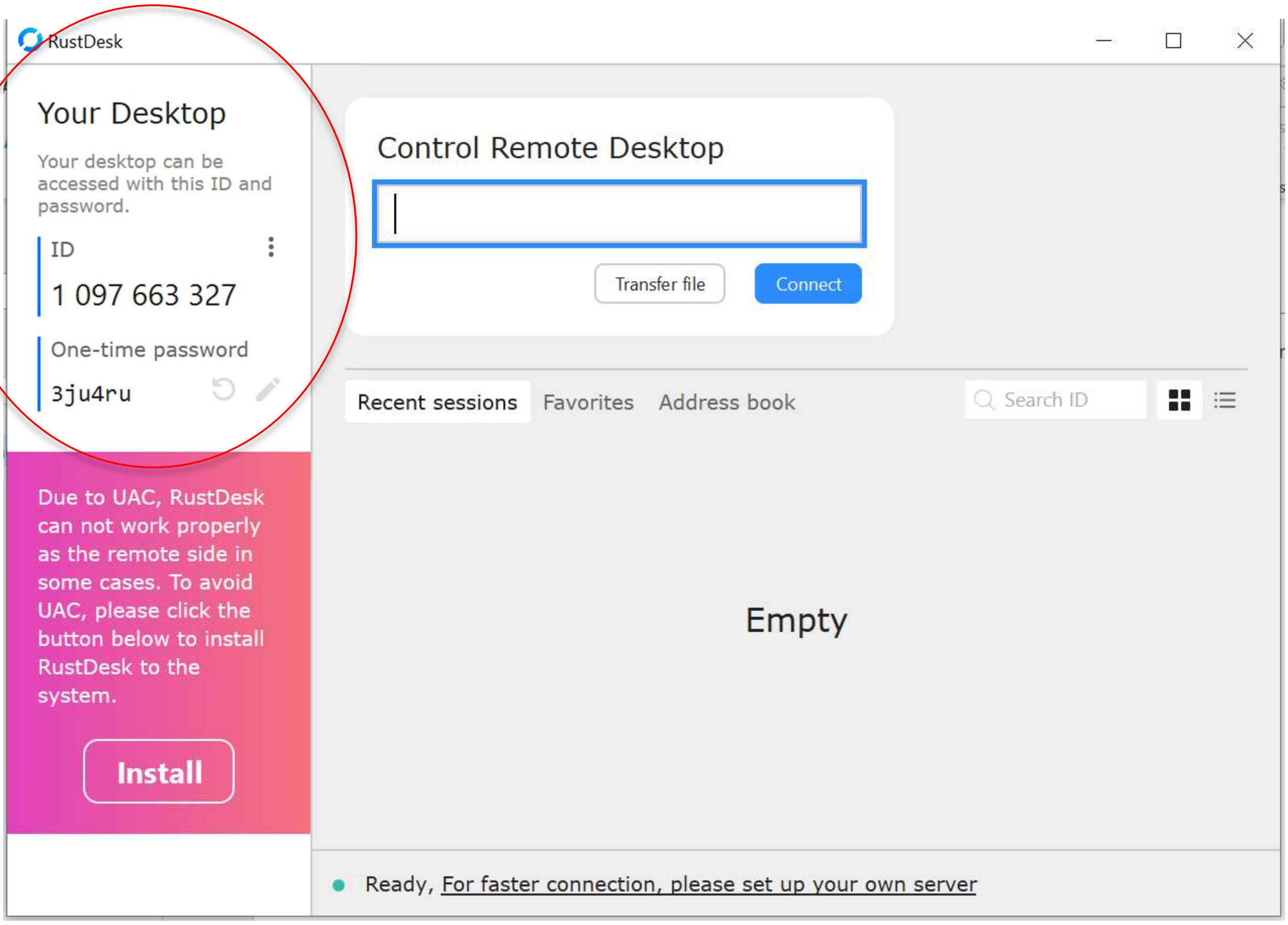
You can then search on the laptop for the name of the phone hotspot "SSID"

Connect to this and this will provide the laptop with an internet connection hosted from the mobile phone.



Please follow the below guidelines to allow the remote access via RustDesk once above steps are done.

Once the program is executed, please provide Nomad Project Engineers with the below credentials to allow the remote connection to the Laptop.



Once provided Nomad Project engineers will then be able to remotely access the test laptop and in turn the CCU once the console cables are connected, see section 3.1 for details.